

Mary Anne Panoyan, M.Sc.

Ph.D. Candidate, Collaborative Specialization in Global Health
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Professional Summary

My research explores the genetic architecture of complex human traits, focusing on brain health and chronic disease, using large-scale genomic datasets. I am interested in applying statistical genetics approaches to investigate how genetic variation, biological mechanisms, and modifiable risk factors contribute to disease risk and to better understand the biological pathways underlying human health and disease. My Research interests broadly include complex trait genomics, human population genetics, modifiable risk factors, GWAS, causal estimates, and risk score predictions.

Summary of Skills

Programming: Python, R, Principal component analysis Machine learning (regression, random forests, neural networks), Mendelian randomization, UK biobank, All of Us, Bash, Linux, Git, PLINK, LDSC, GenomicSEM, twoSampleMR

Education

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|--|------------------------------------|
| Ph.D. Biological Anthropology with a Collaborative Specialization in Global Health | 2021 - Expected August 2027 |
| Department of Anthropology Krembil Centre for Neuroinformatics
University of Toronto Toronto, ON | |
| <ul style="list-style-type: none">• Cumulative GPA: 3.9/4.0• Supervisors: Esteban Parra & Daniel Felsky• Thesis: Genomic and Causal Approaches to Understanding Genetic Determinants of Brain Health and Alzheimer's Disease• Relevant coursework: Evolutionary Anthropology, Population Genetics, Molecular Techniques, Bioinformatics, Molecular Evolution, and Biomedical Machine Learning | |
| M.Sc. Biomedical Informatics | 2019 - 2020 |
| Department of Biomedical and Molecular Sciences School of Computing
Queen's University Kingston, ON | |
| <ul style="list-style-type: none">• Cumulative GPA: 3.67/4.0• Supervisor: Amber Simpson• Thesis: Investigating the role of immune microenvironments and quantitative imaging phenotypes in pancreatic cancer• Relevant coursework: Processing Biomedical Data, Biomedical Data Analysis, Data Mining and Applications, Bioinformatics Cancer Research• Award: Best infographic poster | |
| Honours B.Sc. Combined Specialization in Biology and Forensic Science | 2015 - 2019 |
| Laurentian University Sudbury, ON | |
| <ul style="list-style-type: none">• Cumulative GPA: 3.43/4.0 | |

- **Supervisor:** Kabwe Nkongolo
- **Thesis:** Development and characterization of species diagnostic ISSR and SCAR DNA markers for differentiating red maple (*Acer Rubrum*) and silver maple (*A. Saccharinum*)
- **Relevant coursework:** Biochemistry, Biology, Cell Biology, Genetics, Microbiology, DNA Practicum, Population genetics, Psychology, Experimental Methods in Biology, Statistics for Scientists

Research Experience

Doctoral Researcher

2021 - Present

Krembil Center for Neuroinformatics, Center for Addiction and Mental Health | Toronto ON

- Conducted exome-wide association analyses (GWAS) to investigate **genetic determinants of health and disease**, with a **focus on understudied structural variation**, including short tandem repeats (STRs).
- Applied statistical and **genetic epidemiological methods**, including regression analysis, **Mendelian Randomization (MR)**, **Genetic Structural Equation Modeling (GSEM)**, **linkage disequilibrium score regression (LDSC)**, and **post-GWAS analyses** to investigate genetic associations and potential causal relationships.
- Analyzed large-scale human genetic and phenotypic datasets, including **UK Biobank** and **All of Us**, integrating genetic, health, lifestyle, and imaging-derived phenotypes.
- Performed **next-generation sequencing (NGS) data quality control** and analysis using short-read sequencing data and common genomic file formats, including **BAM**, **VCF**, and related sequence/alignment files in **Linux and Bash environments**.
- Developed and executed reproducible analyses in **R** and **Python**, using statistical, genomic, and bioinformatic packages including PLINK, LDSC, GSEM, TwoSampleMR and GWAS analysis tools.

Graduate Researcher

2019-2020

Simpson Lab, School of Computing

Queen's University, Kingston ON

- Developed robust machine learning models (**neural networks & deep learning**) to predict healthcare outcomes for pancreatic cancer patients from CT images and tissue microenvironments using **TensorFlow** and **Keras**
- Applied statistical analysis including Cox regression to perform **survival analysis on clinical data to predict patient prognosis**
- Disseminated findings with a poster presentation at the Imaging Network Ontario Symposium in Toronto, ON

Undergraduate Researcher

2018 - 2019

Nkongolo Lab, School of Natural Sciences

Laurentian University, Sudbury ON

- Strong proficiency in **molecular biology techniques** including protein purification, Sanger Sequencing, aseptic techniques, bacterial culturing, DNA extraction, DNA cloning, PCR, and gel electrophoresis
- Communicated findings at the Ontario Biology Day Conference in London, ON & published in *Genomes*, 2019

Professional Experience

Chief Presiding Officer (CPO)

2022 – Present

Office of the Registrar at University of Toronto, Mississauga | Mississauga ON

- Organized and set up exam rooms, including organizing seating arrangements and handing out question sheets.
- Policy Enforcement through verifying student photo IDs, monitoring the room to prevent academic offences, and managing late-arrival protocols. Handled student illness, technical or physical disruptions, and documented any violations of conduct rules.

Clinical Trial Research Assistant

2021 - 2022

Population Health Research Institute, Hamilton ON

- Lead study operation as it pertains to internal and external study contacts in the conduct of all study activities (such as timelines, training, protocols and budgets) related to clinical trials across 3 hospital sites within Ontario
- Used subject matter expertise to identify critical trends in study for the development and editing for 20 online CRFs for over 60 study participants
- Oversaw data entry by managing data input software (RedCap) ensuring standards were met, clinical trial operated efficiently and ensured data quality

Operations Coordinator

2020 - 2021

Trillium Health Partners, Mississauga ON

- Conducted analysis and Ad hoc reports, performing data-driven and strategic decision-making, including staff unit transfers
- Coordinated 200+ support staff for the COVID revamp support model for operations stabilization phase and workflow validation
- Validated schedules for 200 staff based on priority principles given by the senior leadership team to ensure support was delivered effectively

Undergraduate & Graduate Research Mentoring

Undergraduate Student Mentor

2023 - 2024

Forensic Science program | Mississauga, ON

- Mentored an undergraduate student (Ellie Wan) in their capstone project for FSC485 course, guiding them through the application of data processing pipeline and machine learning algorithms (random forest and linear regression) to analyze forensic DNA profiles
- Developed research project questions and methods, aligning with the objectives outlined in the course curriculum and ensuring relevance to real-world applications
- Taught interpretation of statistical and machine-learning outputs and supported the student in communicating research findings.
- Student is now an M.Sc. in Laboratory and Pathobiology at the University of Toronto

Mentor for Graduate Student

2022 - 2023

Department of Anthropology | Mississauga, ON

- Mentored a master's student (Anjalika Balasuriya) on their research project characterizing ancestry groups in a Canadian Population. Assistance with Sequence alignments and FaSTR for allele calling and DNA profile analysis
- Student is now in MD at the University of Calgary

Teaching Experience

Sessional Course Instructor

Trent University, Master of Forensic Science Program

Fall 2022

FRSC5003 – Research Design and Statistics in Forensic Science

- Independently teach 24 graduate students through weekly 2-hour lectures and 2-hour tutorials.
- Develop and deliver lectures and tutorials demonstrating the application of statistical methods to forensic data using RStudio.
- Develop course materials, including lecture and tutorial presentations, assignments, and grading rubrics.
- Provide instruction and guidance on research design, statistical analysis, and interpretation of forensic data.

Guest Lecturer

Laurentian University, Master of Forensic Science Program

Summer 2024

FORS5056EL – Advanced Topics in Forensic DNA Analysis

Course Instructor: Krista Currie

- Delivered a guest lecture on advanced topics in forensic DNA analysis to graduate students. Date presented: June 12, 2024

Teaching Assistantships

Practical Teaching Assistant | University of Toronto, Department of Anthropology

Summer 2026 & Fall 2026

ANT202 – Biological Anthropology: Human Variation and Adaptation

Course Instructor: Esteban Parra

- Lead weekly practical sessions on human evolution, genetics, and biological variation for two one hour sessions with class size of approximately 30 students.
- Teach fundamental genetics concepts, including Punnett squares, population genetics and pedigree/genealogy analysis.

Practical Teaching Assistant | University of Toronto, Department of Biology

Fall 2023

BIO259 – Introduction to Biological Data

Course Instructor: Alex Nguyen Ba

- Led weekly practical lectures for approximately 30 students for 2-hour sessions.
- Provide hands-on assistance with R programming and biological data analysis.
- Support students in applying computational methods to biological datasets.
- Invigilate and administer term tests for approximately 400 students.

Tutorial & Marking Teaching Assistant | University of Toronto, Forensic Science Program

Winter 2023 & Fall 2023

FSC416 – Population Genetics

Course Instructor: Frank Wendt

- Lead weekly office hours to mentor students and address gaps in genetics and population genetics knowledge.
- Provide tutorial support and assistance with R programming and data analysis.
- Mark student assignments and provide constructive feedback.

Marking Teaching Assistant | University of Toronto, Forensic Science Program**Winter 2023****FSC350 – Forensic Statistics**

Course Instructor: Shelby Scott

- Marked student assignments and provided feedback on statistical analyses and interpretation for approximately 40 students.

Marking Teaching Assistant | University of Toronto, Forensic Science Program**Summer 2022 & Fall 2024****FSC100 – The Real CSI**

Course Instructor: Shelby Scott

- Provided timely feedback on assignments and essays for approximately 100 students.
- Assisted with development of course materials, including quiz, midterm, and assignment questions.
- Maintained familiarity with weekly lecture materials and readings to support assessment and student learning.
- Invigilated and administered midterm and final examinations.

Marking Teaching Assistant | University of Toronto, Forensic Science Program**Fall/Winter 2022****FSC482 – Professional Practice in Forensic Science**

Course Instructors: Tracy Rogers & Karen Woodall

- Assist with weekly course updates and assessment activities.
- Mark assignments and provide feedback to approximately 80 students.
- Support students in developing professional skills relevant to forensic science.

Practical & Marking Teaching Assistant | University of Toronto, Forensic Science Program**Fall 2022****FSC316 – Forensic Anatomy**

Course Instructor: Lilianna Watamaniuk

- Mark weekly laboratory assignments for approximately 40 students.
- Assist with practical demonstrations and lectures.
- Invigilate and administer midterm and final examinations.

Head Teaching Assistant | University of Toronto, Forensic Science Program**Fall/Winter 2021 - 2024****FSC239 – Introduction to Forensic Science**

Course Instructor: Vivienne Luk, Frank Wendt, Rasmus Larson

- Develop assessment materials, including test questions designed to assess and engage students across course topics.
- Serve as a liaison between students and the course instructor, communicating student concerns and course expectations through email, Microsoft Teams, and Zoom.
- Conduct virtual office hours to mentor students, address academic concerns, and support understanding of course material.
- Provide guidance to students navigating university coursework and academic life.

FSC315 – Forensic Biology

Course Instructor: Nicole Novroski

- Independently managed weekly 3-hour interactive laboratory sessions for 8 students.
- Delivered lectures and demonstrations covering forensic serology, DNA extraction, amplification, quantification, and interpretation.
- Invigilated and administered laboratory examinations and practical assessments.

Community Engagement & Professional Services

UTM Graduate Student Representative**June 2026 - Present**

Sexual Violence Prevention Student Advisory Committee | Mississauga, ON

- Member of the Student Advisory Committee, meeting quarterly to serve as a consultative body to provide feedback, insights, and create initiatives that enhance the Sexual Violence Prevention Centre's impact.

Volunteer**2022 - Present**

Canadian Girls in Science (CAGIS) | Mississauga, ON

- Organized and facilitated monthly events tailored to engage girls, nonbinary, and gender non-conforming youth aged 7-16, fostering interest and participation in science, technology, engineering, and mathematics fields
- Developed and implemented a workshop focused on creating machine learning models using Teachable Machine and Scratch platforms, enabling participants to learn about and apply machine learning concepts in a hands-on, interactive environment

Social Coordinator**2022 - 2023**

Anthropology Graduate Student Union | Toronto, ON

- Organized and executed bi-monthly social events for the Anthropology Graduate Student Union, such as trivia nights, fostering a sense of community and camaraderie among graduate students in the department
- Coordinated logistics and venue arrangements for each event, ensuring smooth execution and adherence to budgetary constraints while maximizing attendance and engagement

Peer Mentor**2022 - 2023**

Anthropology Graduate Student Union | Toronto, ON

- Shared insights and experiences from personal academic journey, including strategies for managing coursework, navigating departmental procedures, and accessing academic resources, to empower the mentee and enhance their academic success.

Peer Review Publications

Published

1. **Panoyan MA.**, & Wendt F. (2023). The role of tandem repeat expansions in brain disorders. Emerging Topics in Life Sciences. doi: 10.1042/ETLS20230022 (**Impact Factor: 3.3**)

2. Shadoff R, **Panoyan MA.**, & Novroski N. (2022). Microbial Forensics: A present to future perspective on genomic targets, bioinformatic challenges, and applications. *Forensic Genomics*, 2(2), 42–64. doi: 10.1089/forensic.2022.0007 (**Impact Factor: 0.37**)
3. **Panoyan MA***, Boyd M*, Michael P, & Nkongolo KK. 2019. Development and characterization of species diagnostic ISSR and SCAR DNA markers for differentiating red maple (*Acer rubrum*) and silver maple (*A. saccharinum*). *Genome* 62(8):527-535. doi: 10.1139/gen-2019-0037 * co-first authors (**Impact Factor: 2**)

Under Review

1. **Panoyan MA.**, Shi Y., Abbatangelo C.L., Adler N., Moo-Choy A., Parra E., ... & Felsky D. (2023). Exome-wide tandem repeats confer large effects on subcortical volumes in UK Biobank participants. medRxiv. <https://doi.org/10.1101/2023.12.11.23299818> (**Molecular Psychiatry, Impact Factor: 10.4**)

In Preparation

2. **Panoyan MA.**, Wendt FR., Parra EJ., & Felsky D. (2026). Investigating the relationship between subcortical brain volume and alcohol consumption using exome-wide tandem repeats.
3. **Panoyan MA.**, Shi Y., Polimanti R., & Wendt FR. (2026). Detecting tandem repeats associated with alcohol use and abuse.

Conferences

Oral Presentations

1. **Panoyan MA**, et al. (2026). *Investigating the causal relationship between cardiometabolic traits and Alzheimer's disease*. Novo Nordisk Healthy Population Discovery Summit, Mississauga, ON, Canada. June 2026.
2. **Panoyan MA**, et al. (2023). *Investigating the role of exome-wide tandem repeats in alcohol use and misuse*. World Congress of Psychiatric Genetics (WCPG), Montreal, QC, Canada. October 2023.
3. **Panoyan MA**, **Boyd M**, Michael P, & Nkongolo KK. (2019). *Development and characterization of species-diagnostic ISSR and SCAR DNA markers for differentiating red maple (*Acer rubrum*) and silver maple (*A. saccharinum*)*. Ontario Biology Day Student Conference, London, ON, Canada. March 2019

Poster Presentations

1. **Panoyan MA**, et al. (2024). *Investigating the relationship between subcortical brain volume and alcohol consumption using exome-wide tandem repeats*. World Congress of Psychiatric Genetics (WCPG), Singapore. October 2024.
2. **Panoyan MA**, et al. (2020). *Investigating the role of immune environments and quantitative imaging phenotypes in pancreatic cancer*. International Molecular and Nanomedicine Organization (IMNO) Conference, Virtual. March 2020.
3. **Panoyan MA**, et al. (2019). *Investigating the role of immune environments and quantitative imaging phenotypes in pancreatic cancer*. CSearch Conference, Queen's University, Kingston, ON, Canada. October 2019.

Competitions

1. **Panoyan MA** & Novroski N. (2022). *Classification of ancestry by supervised machine learning models on forensically relevant SNPs*. ISHI Three-Minute Thesis Competition, Virtual. June 2022. [Video presentation](#).

Grants

Doctoral Completion Award \$9,100	2026
Ontario graduate scholarship \$15,000	2024
Novo Nordisk Network for Healthy Populations Graduate Award \$20,000	2024
SGS Conference Grant \$1,230	2024
Pilot Research Award \$4,000	2023
GEF Graduate Fund Award \$650	2022

Reviewer Activities

Reviewer Translational Psychiatry	Winter 2024
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Professional Memberships

Women's Health Research Cluster Member	2024 - Present
Novo Nordisk Network for Healthy Populations Member	2024 - Present
International Society of Psychiatric Genetics (ISPG) Student Member	2023 - Present

Certifications

• Course Design Teaching Assistant Training Program (TATP), University of Toronto	In Progress
• Educational Technology Teaching Assistant Training Program (TATP), University of Toronto	In Progress
• Accessible and Inclusive Teaching Teaching Assistant Training Program (TATP), University of Toronto	In Progress
• Foundations of Teaching in Higher Education Teaching Assistant Training Program (TATP), University of Toronto	In Progress
• Building a Culture of Consent at the University of Toronto: Consent, Communication, and Sexual Violence Prevention Education Module for Students Sexual Violence Prevention & Support Centre (SVPSC), University of Toronto	2026
• CITI Canada Privacy Course	2021
• Good Clinical Practice (GCP)	2021
• Health Canada Division 5 Drugs for Clinical Trials Involving Humans	2021
• Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans Course on Research Ethics (TCPS 2: CORE)	2021
• Forensic Identification Certificate	2019